Project Review, Repository Guide and Handover Notes

BIN381 Milestone 4 - SDHSA: Unemployment and Labour Market Exclusion

Part A. Project review

What worked

- **Starting with the problem, not the data.** Milestone 1 named one decision problem and five success criteria. Every later choice (which files, which target, which dashboard pages) could be checked against them.
- **Testing the data relationships.** Milestone 1 warned about silent join failures and the rotating panel. Milestone 2 tested the D01-to-D11 join directly and found zero matches. Because we tested it, we built two clean tracks instead of an invented merge.
- **Weighting.** Using the survey weights meant our rates matched Stats SA to within 0.1 percentage point in all four quarters. That one check made the rest of the findings credible.
- **Judging the model on more than accuracy.** Looking at recall, precision and the threshold curve showed that three of four models scored 81% accuracy while finding almost no unemployed people.
- **Splitting the work by CRISP-DM phase** (modelling, evaluation and ethics, deployment in Milestone 3; dashboard, data and modelling, and report and integration in Milestone 4) gave each member a clear deliverable.
- **Reconciling two views of the data.** The dashboard counts records and the R analysis uses survey weights. Putting both side by side (final report, Table 4) showed they agree on patterns and differ on levels by 0.3 to 0.7 points, which made the dashboard defensible.

What surprised us

- A join that returns nothing is a result, not an error.
- The sentinel value 1.8e308 was in up to 87% of some columns and would have wrecked every average.
- The province field had 33 spellings of nine provinces, which produced 33 dummy columns in the first modelling recipe.
- The default 0.50 cut-off made a model with AUC 0.787 look useless.
- Fairness gaps were large (46 points by population group) even though the gender gap was almost zero.
- The formal/informal labels looked reversed when we checked average earnings.

What changed across milestones

Milestone	Original plan	What actually happened
1	Descriptive analysis only; modelling	Kept descriptive focus; recorded risks that later became real
2	Join D11 with D01-D03	Proved this was impossible; two parallel tracks
3	Pick the most accurate model	Chose by recall and AUC after the accuracy trap; added threshold 0.20 and fairness analysis
4	Dashboard and report	built a one-page dashboard plus a Models page; the report adds weighted rates and a headcount view alongside the sample view;

Milestone	Original plan	What actually happened
		flagged the formal/informal issue instead of using it

What we would improve

1. Check category codes against the official QLFS codebook in Milestone 1, not at the end.
2. Share one cleaning function from the start so that Province spellings were fixed once, not twice.
3. Test the dashboard and the app with a real DEL analyst.
4. Version-control the project (Git) instead of passing zip files.
5. Write the model card from the code that actually produced the model, so the two always agree (see known issues).
6. Agree early on who owns monitoring after submission.

What we learned

- **Power BI and insights**

  Building the dashboard taught us that a chart can look convincing and still be misleading. Our dashboard counts records in the sample, not people, so its unemployment rate is 32.9% against 32.6% when the survey weights are applied. The patterns agree, but we now know to label sample counts clearly and not quote them as national figures. I also learned to check what a filter is doing. The youth trend of about 46% only appears when the age slider is set to 15 to 34, so it is not the national rate. Unemployment rate and number of unemployed people are different things: Gauteng and KwaZulu-Natal have the most unemployed people, but North West and Eastern Cape have the highest rates. Finally, we learned that the quarterly chart has to be sorted by quarter, or it looks like a steady decline that isn't real. Next time we would test the dashboard against the cleaned data before writing the findings, not after.

- **Data, R and modelling**

  The biggest lesson for us was how much of the work is in the data preparation. Reading the 18-digit IDs as numbers lost digits, the 1.8e308 values stood in for "not applicable", and 33 spellings of nine provinces would have broken every result if we had missed them. Leaving Quarter out of the duplicate key dropped most of Q2 to Q4, which our rows-kept chart shows. Testing the join instead of assuming it showed that the two data tracks share no person IDs, so we ran them separately. On modelling, we learned the accuracy trap: at the default 0.50 cut-off the random forest was about 81% accurate but found only 6% of unemployed people. Moving the threshold to 0.20 raised recall to 80%, at the cost of many false alarms, so we now judge models on recall, precision and AUC together. Checking performance by group also showed real gaps, with recall of 83% for Black African people against 36% for White people. On deployment, we learned that a 29-row test only proves the app opens, because the drift checks need at least 500 rows.

- **Report, ethics and integration**

  Merging the work taught us that a project only holds together if the numbers agree across the report, the dashboard and the app. The accuracy trap changed how we read any model score, and the fairness gaps (83% recall for Black African people against 36% for White people) have to be stated plainly in the results. We also learned to be honest about what we couldn't do: the failed join between the two data tracks and the doubtful formal and informal coding are in the report, not hidden. Running the Shiny app and its monitoring tab myself showed us that deployment means validating inputs, watching for drift and having a person review every flag.

Part B. How to reproduce

Software. R 4.3 or newer and RStudio. Power BI Desktop (Windows) for the dashboard.

Packages (install once).

```
install.packages(c("tidyverse", "recipes", "rsample", "caret", "ranger", "e1071", "pROC",
                   "randomForest", "C50", "gmodels", "janitor", "knitr", "scales",
                   "shiny", "DT", "ggplot2", "dplyr", "stringr", "rmarkdown"))
```

Run order (open `Bin381-Milestone3.Rproj` first so the working directory is the project root; the seed is 2026 throughout).

Step	Run	Produces	Expected check
1	Knit the Milestone 2 notebook (if raw data is present)	`data/Cleaned/*`	170,377 and 45,322 rows exported
2	`source("02b_fix_baked.R")`	`train_baked_fixed.rds`, `test_baked_fixed.rds`, `recipe_prepped_fixed.rds`	Province dummy columns: 9 provinces
3	Knit `Milestone3.Rmd`, then `source("03_evaluation.R")`, `03b_rf_balanced.R`, `03c_finalise.R`	`outputs/*`, `final_model.rds`, `final_model_metadata.rds`	Final model recall 0.800, AUC 0.787
4	`source("04_build_powerbi_data.R")`	`powerbi/*.csv`	Printed quarterly rates 0.329, 0.335, 0.321, 0.319
5	Knit `Milestone4_Final_Analysis.Rmd`	HTML report of the final numbers	Matches report section 3
6	`source("deployment/tests/smoke_test.R")`	console	"All smoke tests passed."
7	`shiny::runApp("deployment")`	the app	Opens with four tabs
8	Open `Dashboard.pbix`, **Refresh**	dashboard	Sample figures: pooled rate 32.9%, quarters 33.1, 33.8, 32.4, 32.2; youth NEET 42.1%

Part C. Monitoring, maintenance and handover notes

Who owns what

Role	Responsibility
Model owner	Decides on retraining and retirement; signs off monitoring results

Role	Responsibility
Data owner	Supplies each new quarterly batch and checks the data source for changes
Analyst	Runs the Monitoring tab, reads the log, escalates ACTION rows
Ethics contact	Reviews fairness gaps each year and any change in how flags are used

Quarterly routine (about 30 minutes)

1. Get the new QLFS batch in the same layout as the scoring template.
2. Open the app, Monitoring tab, upload the batch, set a window label (for example 2025-Q1).
3. Read the table. Colours: green OK, amber Watch, red ACTION.
4. Save a copy of `monitoring_log.csv` with the date.
5. Refresh the Power BI data if the cleaned panel has been updated (and re-run `04_build_powerbi_data.R` if the weighted tables are used).

What to do when a check fires

Alert	First response	Escalate if
Score drift (PSI above 0.20)	Check for a change in the source data or in code definitions	Persists in the next batch
Feature drift (KS p below 0.01 and D at least 0.10)	Find which field moved; check coding	The field is a top predictor (age, education, province)
Missing data above 5%	Contact the data owner	Not fixed before the next run
Recall drops more than 10 points from 0.80 (needs real outcomes)	Retrain with the newest labelled QLFS data	Always notify the model owner
Group recall gap widens by more than 10 points	Retrain with reweighting; review use for that group	Always notify the ethics contact
Recall below 0.60 in two consecutive windows	Retire and rebuild	Immediately

Retraining steps

1. Add the newest QLFS waves to the raw data folder and re-run the preparation notebook.
2. Re-run `02b_fix_baked.R` and the Milestone 3 scripts.
3. Compare the new model with the old one on the new hold-out data (AUC, recall at the chosen threshold, subgroup recall).
4. Re-run `01_build_monitoring_baseline.R` so that the monitoring reference matches the new model.
5. Update the model card and the dashboard numbers table, and record the change in the AI and ethics log.